Exercise Sheet

Lecture 5

Eigenvalues and Eigenvectors

Throughout, V is a nonzero finite-dimensional vector space, $T \in \mathcal{L}(V)$ is an operator, and $E(\lambda, T) = \ker(T - \lambda I)$ is its eigenspace. Tags indicate the flavour of each problem; a $\star$ marks a harder one.

Core tutorial route

Work **5.1(i), 5.2, 5.6, 5.10, 5.14, and 5.16** in that order. This route covers finding eigenvalues and eigenspaces, the check by trace and determinant, real and complex eigenvalues, the eigenvalues of an invertible operator, the eigenspace decomposition, invariant subspaces, and finding a basis in which the matrix is upper triangular.

Additional practice: 5.1(ii)–(iii), 5.4, 5.7–5.9, 5.12–5.13. **Bridge:** 5.1(iv), 5.3 (complex eigenvalues and differentiating polynomials). **Extension:** 5.5, 5.11, 5.15 (the matrix whose entries all equal one, the eigenvalues of a product of two operators taken in either order, and a proof that an eigenvalue exists without determinants).

Finding eigenvalues and eigenvectors

Exercise 5.1 (computational) Find the eigenvalues of the matrix A and a basis of each eigenspace, where (i) $A = \left(\begin{smallmatrix} 4 & -5 \\ 2 & -3 \end{smallmatrix}\right)$; (ii) $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$; (iii) $A = \left(\begin{smallmatrix} 0 & 1 \\ -6 & 5 \end{smallmatrix}\right)$; (iv) $A = \left(\begin{smallmatrix} 1 & -2 \\ 1 & -1 \end{smallmatrix}\right)$ (work over $\mathbb{C}$). Check the eigenvalues you find against $\operatorname{tr} A$ and $\det A$.

Answer. (i) Eigenvalues 2 and -1; $E(2, A) = \operatorname{span}\big((5, 2)\big)$, $E(-1, A) = \operatorname{span}\big((1, 1)\big)$. (ii) Eigenvalues 3 and -1; $E(3, A) = \operatorname{span}\big((1, 1)\big)$, $E(-1, A) = \operatorname{span}\big((1, -1)\big)$. (iii) Eigenvalues 2 and 3; $E(2, A) = \operatorname{span}\big((1, 2)\big)$, $E(3, A) = \operatorname{span}\big((1, 3)\big)$. (iv) Eigenvalues $\pm i$; $E(i, A) = \operatorname{span}\big((1 + i, 1)\big)$, $E(-i, A) = \operatorname{span}\big((1 - i, 1)\big)$.

Exercise 5.2 (computational) Compute the eigenvalues of the rotation matrix $\left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right)$ over $\mathbb{C}$. For which values of α are they real?

Answer. $\lambda = e^{\pm i\alpha}$; real if and only if $\sin\alpha = 0$, i.e. $\alpha \in \pi\mathbb{Z}$ (in $[0, 2\pi)$ only $\alpha = 0$ and $\alpha = \pi$).

Exercise 5.3 (computational) Let $D \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be the differentiation operator. Find all eigenvalues and eigenvectors of D.

Answer. The only eigenvalue is 0; $E(0, D) = \operatorname{span}(1)$ (the geometric multiplicity 1 is less than the algebraic multiplicity 3, so D is defective).

Exercise 5.4 (computational) Find the eigenvalues of the matrix A, their algebraic and geometric multiplicities, and a basis of each eigenspace. Which of these matrices are defective?
(i) $A = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{smallmatrix}\right)$; (ii) $A = \left(\begin{smallmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{smallmatrix}\right)$; (iii) $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 3 \end{smallmatrix}\right)$; (iv) $A = \left(\begin{smallmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{smallmatrix}\right)$; (v) $A = \left(\begin{smallmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{smallmatrix}\right)$.

Answer. Both multiplicities are given for the repeated eigenvalues; for the others both equal 1. (i) 1, 2, 4; $E(1, A) = \operatorname{span}\big((-3, 1, 1)\big)$, $E(2, A) = \operatorname{span}\big((0, 1, -1)\big)$, $E(4, A) = \operatorname{span}\big((0, 1, 1)\big)$; not defective. (ii) 3 (algebraic 2, geometric 2) and 5; $E(3, A) = \operatorname{span}\big((0, 1, 0), (-1, 0, 1)\big)$, $E(5, A) = \operatorname{span}\big((1, 2, 1)\big)$; not defective. (iii) 2 (algebraic 2, geometric 1) and 3; $E(2, A) = \operatorname{span}\big((1, 0, -1)\big)$, $E(3, A) = \operatorname{span}\big((0, 0, 1)\big)$; defective. (iv) 3 and -1 (algebraic 3, geometric 2); $E(3, A) = \operatorname{span}\big((1, 1, 0, 0)\big)$, $E(-1, A) = \operatorname{span}\big((1, -1, 0, 0), (0, 0, 1, 0)\big)$; defective. (v) 3, 1

and -1 (algebraic 2, geometric 2); $E(3,A) = \operatorname{span}\big((1,1,0,0)\big)$, $E(1,A) = \operatorname{span}\big((0,0,1,-1)\big)$, $E(-1,A) = \operatorname{span}\big((1,-1,0,0),(0,0,1,1)\big)$; not defective.

Hint. (i)–(iii) Expand the determinant along a row or column with the most zeros. (iv)–(v) For a block diagonal matrix $A = \left(\begin{smallmatrix} B & 0 \\ 0 & C \end{smallmatrix}\right)$ we have $\chi_A = \chi_B\,\chi_C$; pad the eigenvectors of each block with zeros.

Exercise 5.5 ($\star$) Let J be the $n \times n$ real matrix ($n \geq 2$) with every entry equal to 1. Find all eigenvalues of J, their algebraic and geometric multiplicities, and a basis of each eigenspace.

Hint. The matrix J has rank 1. Look at how J acts on the vector $(1,\dots,1)$ and on the vectors whose coordinates sum to 0.

Properties of eigenvalues

Exercise 5.6 (proof)

(i) Prove that T is invertible if and only if 0 is not an eigenvalue of T.

(ii) Suppose T is invertible and λ is an eigenvalue of T. Prove that $\lambda \neq 0$ and that $1/\lambda$ is an eigenvalue of T^{-1} with the same eigenvectors.

Hint. (i) The number 0 is an eigenvalue exactly when $\ker T \neq \{0\}$. On a finite-dimensional space an operator is injective if and only if it is invertible. (ii) Apply T^{-1} to both sides of $Tv = \lambda v$. To get the eigenspaces equal rather than merely one inside the other, run the same step backwards.

Exercise 5.7 (computational) Find the eigenvalues (i) of the matrix $A^2 - A$, where $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right)$; (ii) of the matrix A^{-1}, where $A = \left(\begin{smallmatrix} 2 & 0 \\ 1 & 3 \end{smallmatrix}\right)$, together with an eigenvector for each of them.

Answer. (i) 0 and 6. (ii) Eigenvalues $\frac{1}{2}$ (eigenvector $(1,-1)$) and $\frac{1}{3}$ (eigenvector $(0,1)$).

Exercise 5.8 (computational) Verify that the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is an involution, i.e. $A^2 = I$, and find the eigenspaces $E(1,A)$ and $E(-1,A)$.

Answer. $A^2 = I$; $E(1,A) = \operatorname{span}\big((1,1)\big)$, $E(-1,A) = \operatorname{span}\big((1,-1)\big)$.

Exercise 5.9 (proof)

(i) For a polynomial $p(t) = a_0 + a_1 t + \cdots + a_m t^m$ we write $p(T) = a_0 I + a_1 T + \cdots + a_m T^m$, where T^k is the composition repeated k times and $T^0 = I$. Suppose λ is an eigenvalue of T and $p \in \mathcal{P}(\mathbb{F})$. Prove that $p(\lambda)$ is an eigenvalue of $p(T)$ with the same eigenvector.

(ii) Let $c \in \mathbb{F}$. Prove that the operators T and $T + cI$ have exactly the same eigenvectors, and that λ is an eigenvalue of T if and only if $\lambda + c$ is an eigenvalue of $T + cI$.

Hint. (i) First show by induction on k that $T^k v = \lambda^k v$, then apply $p(T) = \sum_k a_k T^k$ to v term by term. (ii) One line suffices for each direction: $Tv = \lambda v$ gives $(T + cI)v = (\lambda + c)v$, and $(T + cI)v = \mu v$ gives $Tv = (\mu - c)v$.

Exercise 5.10 (proof) Suppose $T^2 = I$, i.e. T is an involution. Prove that every eigenvalue of T is ± 1 and that $V = E(1,T) \oplus E(-1,T)$.

Hint. For the eigenvalues, apply T twice to an eigenvector. For the decomposition, split v into a part that T fixes and a part that T negates.

Exercise 5.11 ($\star$) Let $S, T \in \mathcal{L}(V)$. Prove that the operators ST and TS have the same eigenvalues.

Hint. If $STv = \lambda v$ with $\lambda \neq 0$, look at how TS acts on Tv. Treat the case $\lambda = 0$ separately through invertibility.

Exercise 5.12 (proof) An operator T is called *nilpotent* if $T^k = 0$ for some $k \geq 1$. Prove that the only eigenvalue of a nilpotent operator is 0.

Hint. Apply the equation $T^k = 0$ to an eigenvector: on one side $T^k v = \lambda^k v$, on the other it is zero, and $v \neq 0$.

Upper triangular form

Exercise 5.13 (computational)

(i) Read off the eigenvalues of the matrix $A = \begin{pmatrix} 5 & 2 & 7 \\ 0 & 5 & -1 \\ 0 & 0 & 3 \end{pmatrix}$ and state its trace and determinant. Is the matrix A invertible?

(ii) Without computing any eigenvectors, decide whether the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ has a basis of eigenvectors in $\mathbb{R}^3$.

Answer. (i) Eigenvalues 5 (algebraic multiplicity 2) and 3; $\operatorname{tr} A = 13$, $\det A = 75$; invertible. (ii) Yes: its eigenvalues 1, 4 and 6 are distinct.

Hint. (ii) Read the eigenvalues off the diagonal and compare their number with $\dim \mathbb{R}^3$.

Exercise 5.14 (computational) For the matrix $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ determine which of the following subspaces are invariant:

$$U_1 = \operatorname{span}((1,0,0)), \qquad U_2 = \operatorname{span}((1,1,0)),$$
$$U_3 = \operatorname{span}((1,0,0),(0,1,0)), \qquad U_4 = \operatorname{span}((0,0,1)).$$

Which of the invariant subspaces is not an eigenline?

Answer. U_1, U_3, U_4 are invariant; U_2 is not. The plane U_3 is invariant but is not an eigenline.

Hint. Test the spanning vectors: a subspace is invariant when the image of every spanning vector lies in it. One of the four subspaces is a plane, not a line.

Exercise 5.15 (proof) Let V be a nonzero n-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove without determinants that T has an eigenvalue. Choose $v \neq 0$ and explain why the vectors $v, Tv, \dots, T^n v$ are linearly dependent, why the resulting polynomial is nonconstant, where the factorization over $\mathbb{C}$ is needed, and why at least one linear factor has a nonzero kernel.

Hint. Count first: $n+1$ vectors in an n-dimensional space must be linearly dependent. Factor the resulting polynomial over $\mathbb{C}$, then ask what would follow if every factor $T - \lambda_j I$ were injective.

Exercise 5.16 (computational) Let $T \in \mathcal{L}(\mathbb{C}^2)$ be given by $Tx = Ax$ with $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Find an ordered basis v_1, v_2 in which the matrix $\mathcal{M}(T, (v_1, v_2))$ is upper triangular, and check directly that this matrix is triangular.

Answer. For example, $v_1 = (-i, 1)$, $v_2 = (1, 0)$ give $\mathcal{M}(T, (v_1, v_2)) = \begin{pmatrix} -i & 1 \\ 0 & i \end{pmatrix}$.

Hint. The first column of an upper triangular matrix is $(\lambda_1, 0)^{\mathsf{T}}$, i.e. $Tv_1 = \lambda_1 v_1$. Start from an eigenvector.